The H6N1 influenza A virus is an enveloped RNA virus with potential for zoonosis (transmission from animals to humans). It obtains its plasma membrane from its host cell, but the composition of the viral envelope and vertebrate plasma membrane are distinct. The viral envelope consists mostly of phosphatidylethanolamine as well as some negatively charged glycerophospholipids. The outer membrane contains hemagglutinin (HA) proteins that bind to the carbohydrate known as sialic acid (SA) on host cell receptor proteins, facilitating viral entry. It also contains neuraminidase enzymes that break the bond between SA and galactose in the membrane glycans of various host species, allowing new virions to be released.

Minute changes in the genes that code for HA proteins can alter the virulence, host, and attachment mechanism. In general, influenza A infects avian species by attaching to tracheal tissue, which displays SA groups bound to receptors through an α-2,3-glycosidic linkage to galactose (Figure 1). In contrast, human cells contain α-2,6-glycosidic linkages between SA and galactose; avian viruses generally cannot attack human cells due to poor recognition of these linkages by H6N1 HA proteins.

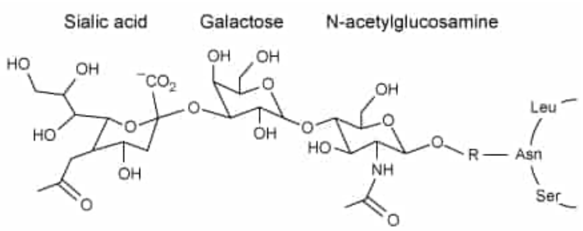


Figure 1 Structure of the influenza A receptor in avian cell membranes

In a recent study, however, a G225D substitution in HA, caused by a GGC to GAU codon mutation, enabled avian virions to infect human cells. These virions are known as the H6-G225D strain. The stability of G225D HA proteins was compared to that of the HA found in the H6-P186L strain, which is created by a single CCC to CUC mutation and exclusively infects avian cells, and to the stability of HA isolated from H1N1 virions, which targets only humans. The viral RNA of H6-P186L, H6-G225D, and H1N1 was cloned using cDNA and expressed in embryonic kidney cells. The melting curve of each purified HA protein was determined using differential scanning calorimetry, with peaks representing melting temperatures (Figure 2).

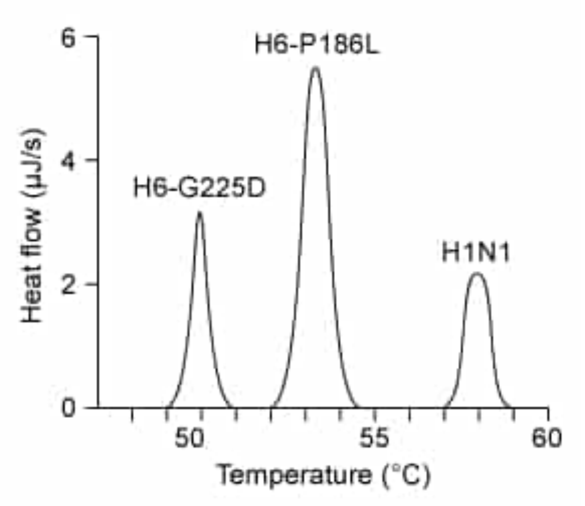


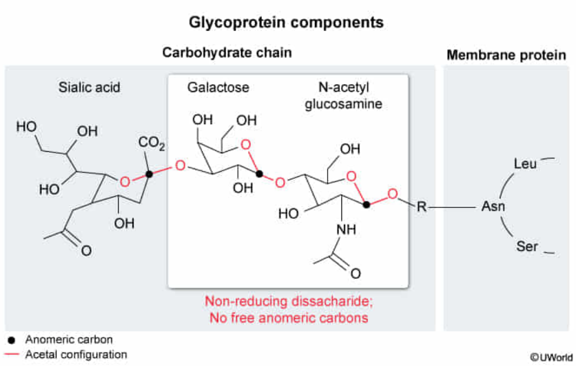
Figure 2 Hemagglutinin melting curves for H6N1 and H1N1 influenza A strains

Which statement best describes the structure of the avian receptor of influenza A shown in Figure 1?

- A. It is a glycoprotein with the anomeric carbon of sialic acid bound to a nonreducing disaccharide [49%]
- B. It is a glycolipid with three reducing sugars bound to each other through their anomeric carbons [10%]
- C. It is a glycoprotein with sialic acid bound to the anomeric carbon of a reducing disaccharide [34%]
- D. It is a glycolipid with three nonreducing sugars in which only one bond contains an anomeric carbon [5%]

Correct
49% answered correctly

Explanation:



Carbohydrates on the outer surface of a cell can act as markers for cell recognition and signaling. During post-translational modification and lipid processing, sugar chains (glycans) are attached to proteins or lipids, converting them into glycoproteins or glycolipids, respectively. These sugars can be classified as reducing if they contain **a free anomeric carbon** that can be oxidized, and as nonreducing when this carbon is linked to another molecule. In linear form, the free anomeric carbon is either an aldehyde or a ketone. In cyclic form, reducing sugars have **hemiacetal** (aldoses) or **hemiketal** (ketoses) configurations at their anomeric carbons whereas nonreducing sugars have acetal or ketal groups.

Figure 1 shows that the influenza A receptor is a **glycoprotein** because the carbohydrate chain is linked to a protein through the amino acid asparagine. The terminal group of the chain consists of SA bound to two, linked sugar molecules (a disaccharide). Both sugars in the disaccharide are bound to another molecule at their anomeric carbons, so neither contains a hemiacetal or hemiketal. Therefore, the disaccharide is a **nonreducing sugar**, and the correct description of the receptor should state that it is a **glycoprotein** with SA bound to a **nonreducing disaccharide**.

(Choices B and D) The influenza A receptor is a *glycoprotein*. Glycolipids have carbohydrates attached to lipids such as fatty acids, glycerol, or sphingosine backbones. The terminal group of the sugar chain contains *three* bonds through anomeric carbons, not just one.

(Choice C) To be a reducing sugar, the disaccharide bound to the anomeric carbon of SA would need to have a free anomeric carbon. Instead, the anomeric carbon is bound to an R group, making it a *nonreducing* sugar.

Educational objective:
Reducing sugars contain free anomeric carbons that provide reducing power when they are oxidized. In linear form the anomeric carbon is an aldehyde or a ketone, and in cyclic form reducing sugars have hemiacetal or hemiketal configurations. Nonreducing sugars contain acetal or ketal structures in their cyclic forms.

Time spent:0 sec
Subject:
Biochemistry

QID:403567
System:
Carbohydrates, Nucleotides, and Lipids

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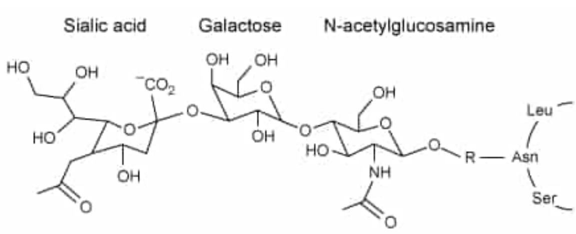


Figure 1 Structure of the influenza A receptor in avian cell membranes

In a recent study, however, a G225D substitution in HA, caused by a GGC to GAU codon mutation, enabled avian virions to infect human cells. These virions are known as the H6-G225D strain. The stability of G225D HA proteins was compared to that of the HA found in the H6-P186L strain, which is created by a single CCC to CUC mutation and exclusively infects avian cells, and to the stability of HA isolated from H1N1 virions, which targets only humans. The viral RNA of H6-P186L, H6-G225D, and H1N1 was cloned using cDNA and expressed in embryonic kidney cells. The melting curve of each purified HA protein was determined using differential scanning calorimetry, with peaks representing melting temperatures (Figure 2).

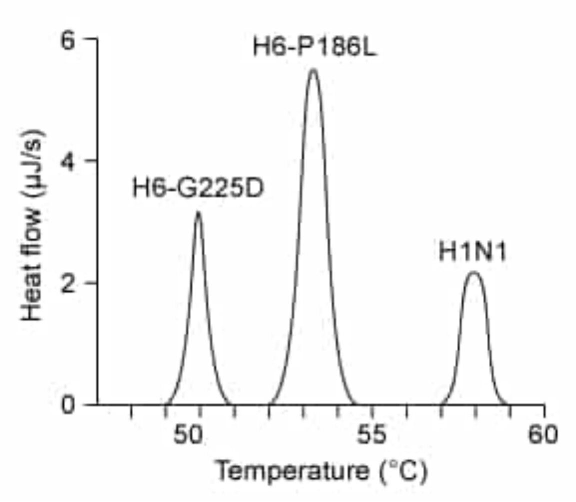


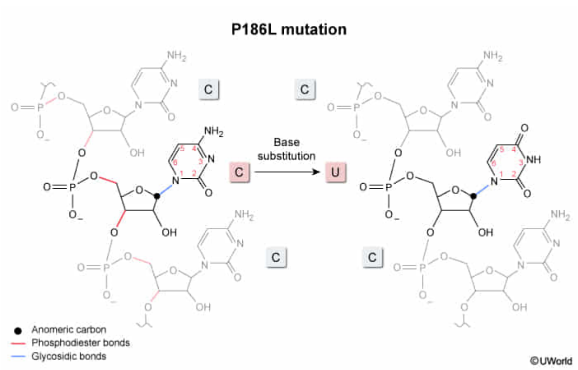
Figure 2 Hemagglutinin melting curves for H6N1 and H1N1 influenza A strains

Compared to the wild-type HA RNA transcript in H6N1, which of the following features is mostly likely found in the H6-P186L RNA transcript?

- A. A carbonyl group in uracil from the wild-type strain is replaced with an amine group at the mutation site [9%]
- B. A glycosidic linkage between a deoxyribose sugar and cytosine in the wild-type transcript is missing at the mutation site [13%]
- C. The H6-P186L transcript contains phosphoanhydride bonds between the 3′ and 5′ carbons of adjacent sugars near the mutation site [10%]
- D. The mutant transcript contains a glycosidic linkage between uracil and ribose instead of cytosine and ribose at the mutation site [66%]

Correct
67% answered correctly

Explanation:



The genetic material of RNA viruses such as influenza A is composed of cytosine, uracil, adenine, and guanine nucleotides bound to each other in single strands. In the backbone of these strands, adjacent ribose sugars are joined by **phosphodiester** bonds at the 5′ and 3′ carbons. Each nucleotide contains a nitrogenous base bound to a sugar through a covalent bond known as a **glycosidic bond**.

According to the passage, the H6-P186L subtype of H6N1 influenza A contains a CCC to CUC mutation in the viral RNA, indicating that a **uracil base** took the place of a **cytosine** base. Therefore, this position in the mutant RNA contains a **ribose** bonded through a glycosidic linkage to uracil instead of cytosine.

(Choice A) Cytosine and uracil are both pyrimidines but differ in the functional groups attached to their aromatic rings. Cytosine has an amino group attached to carbon 4 whereas uracil has a carbonyl at that position, so replacing a carbonyl with an amine would convert uracil to cytosine, not cytosine to uracil.

(Choice B) RNA viruses use ribose, not deoxyribose, in their phosphate-sugar backbones. Nucleotides with deoxyribose sugars are only used to generate DNA strands.

(Choice C) Adjacent nucleotides within RNA and DNA are connected by phosphodiester bonds, not phosphoanhydride bonds. Phosphoanhydride bonds are used to connect the phosphate groups to each other within nucleotide di- and tri-phosphates or to connect the individual nucleotides of specific metabolites like NAD^+ and FAD.

Educational objective:
RNA nucleotides are composed of a ribose sugar with a phosphate group bound to a nitrogenous base (eg, adenine, guanine, cytosine, and uracil). Adjacent sugars in the RNA backbone are bound to each other by phosphodiester bonds, and bases are linked to sugars by glycosidic bonds.

Time spent:0 sec
Subject:
Biochemistry

QID:403568
System:
Carbohydrates, Nucleotides, and Lipids

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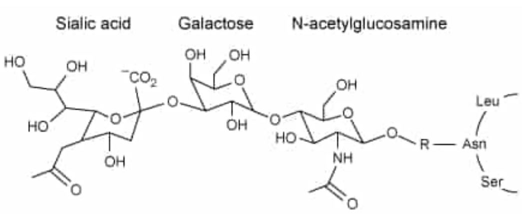


Figure 1 Structure of the influenza A receptor in avian cell membranes

In a recent study, however, a G225D substitution in HA, caused by a GGC to GAU codon mutation, enabled avian virions to infect human cells. These virions are known as the H6-G225D strain. The stability of G225D HA proteins was compared to that of the HA found in the H6-P186L strain, which is created by a single CCC to CUC mutation and exclusively infects avian cells, and to the stability of HA isolated from H1N1 virions, which targets only humans. The viral RNA of H6-P186L, H6-G225D, and H1N1 was cloned using cDNA and expressed in embryonic kidney cells. The melting curve of each purified HA protein was determined using differential scanning calorimetry, with peaks representing melting temperatures (Figure 2).

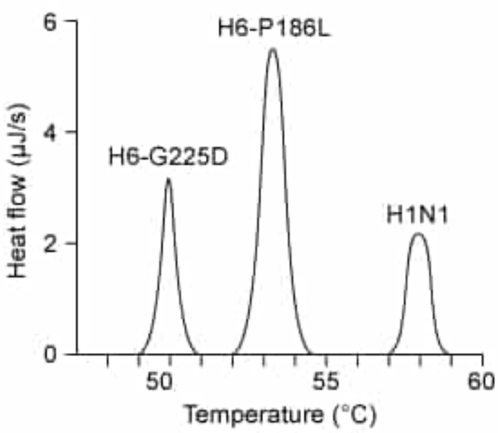


Figure 2 Hemagglutinin melting curves for H6N1 and H1N1 influenza A strains

Based on Figure 2, which of the following is true of H6-G225D hemagglutinin?

A. H6-G225D hemagglutinin is more stable than the hemagglutinin derived from the other virions. [8%]

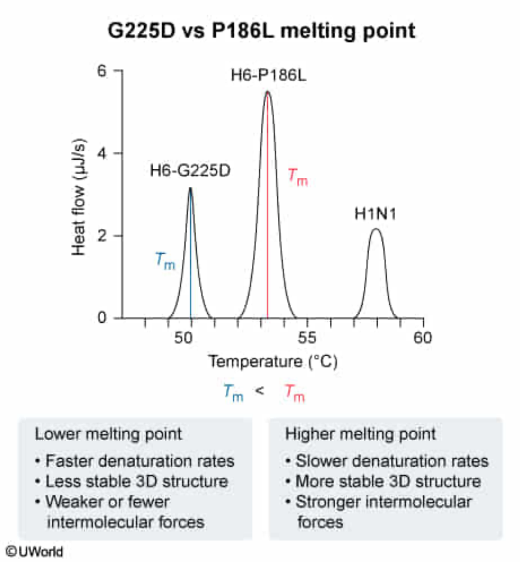
B. H6-G225D hemagglutinin denatures faster than the other hemagglutinin forms tested at any given temperature. [84%]

C. H6-G225D hemagglutinin has a higher melting temperature than the other forms of hemagglutinin. [2%]

D. H6-G225D hemagglutinin experiences stronger intermolecular forces than the other forms of hemagglutinin. [4%]

Correct
84% answered correctly

Explanation:



The stability of a protein can be analyzed by measuring the effect of temperature on its folded three-dimensional structure. Increasing temperature provides **heat energy** to disrupt **intermolecular forces** that promote folding, such as hydrogen bonds and hydrophobic interactions. As a result, proteins exposed to high temperatures lose their secondary, tertiary, and quaternary structures and can no longer fulfill their function in the cell. This process, called denaturation, is used to detect the melting point, at which there are equal amounts of folded and unfolded protein in a sample.

In techniques such as differential scanning calorimetry (DSC), the rate at which proteins denature is correlated with the **melting temperature T_m**, shown as a peak in the DSC curve. Proteins that are more susceptible to heat denature faster at a given temperature and exhibit lower melting temperatures.

Figure 2 shows the melting curves for the HA proteins of three different subtypes of influenza A. The H6-G225D HA protein has the *lowest* melting temperature at 50.2°C **(Choice C)**, and therefore it is the **most susceptible to denaturation**. In contrast, the H6-P186L HA protein and the H1N1 HA protein are more stable with a melting temperatures of 53.5°C and 58.0°C, respectively, so greater amounts of heat energy will be required to disrupt their folded structures. The gap between the T_m of H6-G225D and the other forms of HA tested indicates that within the temperature range shown, the HA of H6-G225D will **denature at a faster rate** than the other forms.

(Choices A and D) Intermolecular forces make proteins more stable and increase their T_m. H6-

G225D melts at a lower temperature than the other forms of HA, so the intermolecular forces in H6-G225D must be *weaker*, making the overall structure *less stable*.

Educational objective:
Increasing temperatures can cause proteins to unfold (denature) by disrupting the intermolecular forces that maintain secondary, tertiary, and quaternary structures. When the melting temperature of a protein is achieved, there are equal amounts of folded and unfolded proteins. A lower melting temperature indicates lower stability and a higher denaturation rate.

Time spent:0 sec	**QID**:403569
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids

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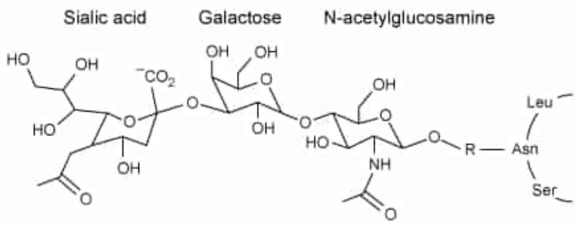


Figure 1 Structure of the influenza A receptor in avian cell membranes

In a recent study, however, a G225D substitution in HA, caused by a GGC to GAU codon mutation, enabled avian virions to infect human cells. These virions are known as the H6-G225D strain. The stability of G225D HA proteins was compared to that of the HA found in the H6-P186L strain, which is created by a single CCC to CUC mutation and exclusively infects avian cells, and to the stability of HA isolated from H1N1 virions, which targets only humans. The viral RNA of H6-P186L, H6-G225D, and H1N1 was cloned using cDNA and expressed in embryonic kidney cells. The melting curve of each purified HA protein was determined using differential scanning calorimetry, with peaks representing melting temperatures (Figure 2).

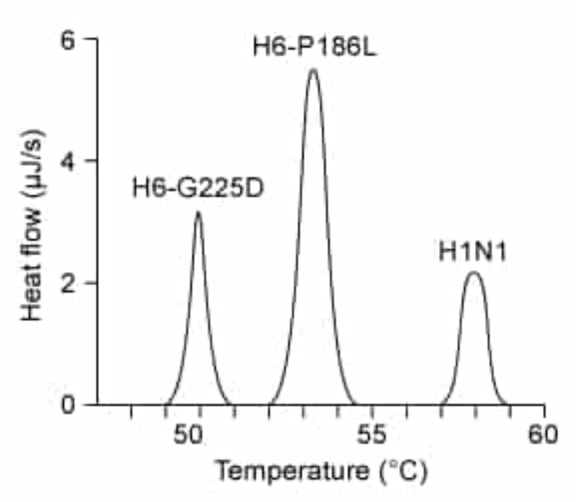


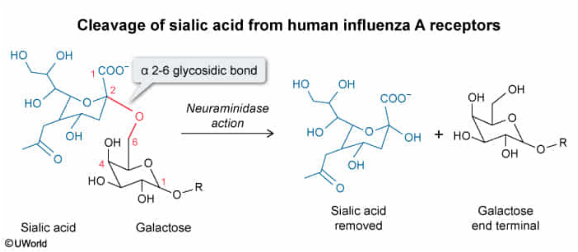
Figure 2 Hemagglutinin melting curves for H6N1 and H1N1 influenza A strains

On human receptors, neuraminidase action would lead to a decrease in what type of bond?

- A. *N*-linked glycosidic bond [12%]
- B. β-1,4-glycosidic bond [13%]
- C. α-2,3-glycosidic bond [23%]
- D. α-2,6-glycosidic bond [50%]

Correct
51% answered correctly

Explanation:



Glycosidic bonds form when a nucleophile, such as a hydroxyl or amino group, attacks the anomeric carbon of a sugar molecule. They are classified based on the numbers assigned to the linked carbons and on whether the anomeric carbon is in the α- or the β-conformation. For example, a bond between carbon 1 of one glucose molecule in the α-conformation and carbon 6 of another is an α-1,6 linkage.

In the host cell receptors, SA is linked to **galactose** through a glycosidic bond. Based on the information in the passage, neuraminidase is an enzyme that **removes SA** from these receptors on the membranes of various host species, and therefore neuraminidase must break the glycosidic bond. According to the passage, avian receptors contain α-2,3-glycosidic bonds whereas human receptors have **α-2,6-glycosidic bonds**. Although the variants mentioned in the passage affect the ability of mutant HA protein to bind specific

SA conformations, neuraminidase activity is unaffected and is further noted to be nonspecific and capable of recognizing the glycans of various host species. Therefore, neuraminidase action in humans is expected to increase the cleavage of *human* conformers of SA and thus reduce the number of α-2,6-glycosidic bonds.

(Choice A) *N*-linked glycosidic bonds form between a sugar molecule and the amine group of an amino acid, usually asparagine, as shown in Figure 1. SA is not directly attached to asparagine, so neuraminidase does not break *N*-linked glycosidic bonds.

(Choice B) A β-1,4-glycosidic bond joins the two sugars in the disaccharide (galactose and *N*-acetylglucosamine) to which SA is linked, but it is not the bond between SA and galactose. Neuraminidase does not cleave this bond.

(Choice C) The question asks for the effect of neuraminidase in *humans* rather than avian species. The α-2,3-glycosidic bond to SA is only present in avian species whereas the glycosidic bonds in humans are in the α-2,6 configuration.

Educational objective:
Glycosidic bonds are created by the linkage of a nucleophilic functional group, such as a hydroxyl or amino group, to the anomeric carbon of a sugar molecule. The bond is classified based on the assigned numbers of the linked carbons and on whether the anomeric carbon is in the α- or the β-conformation.

Time spent:0 sec	**QID**:403570
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

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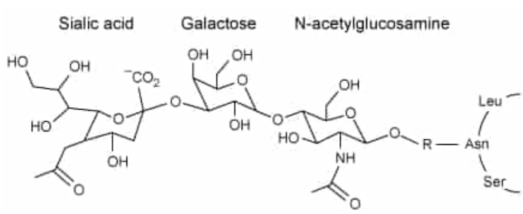

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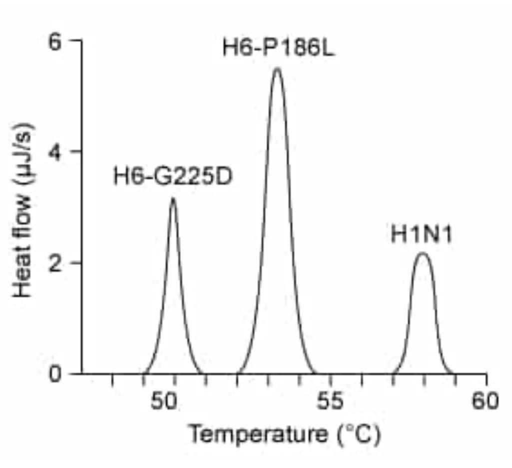


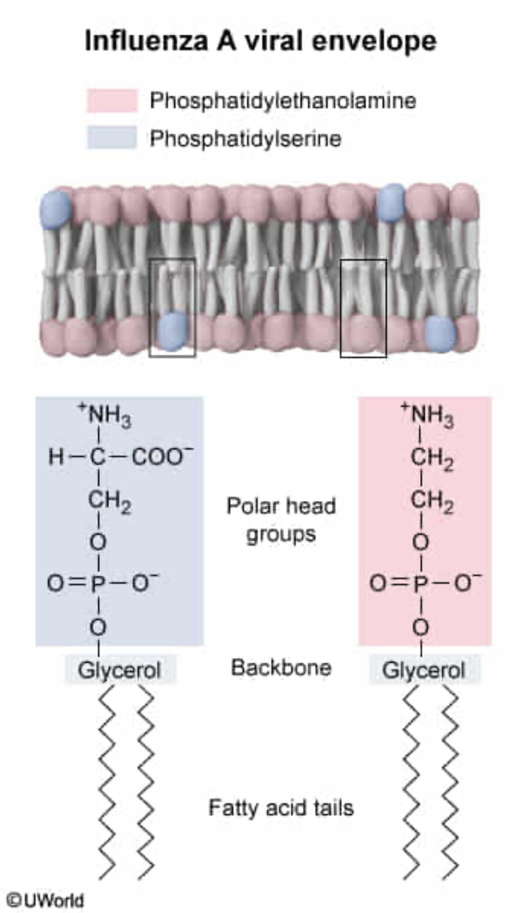
Figure 2 Hemagglutinin melting curves for H6N1 and H1N1 influenza A strains

The absence of which of the following lipids is most likely to affect the assembly of viral envelopes?

A. Palmitic acid [7%]
B. Triacylglycerides [25%]
C. Sphingomyelin [19%]
D. Phosphatidylserine [46%]

Correct
46% answered correctly

Explanation:



The plasma membrane is composed mainly of proteins, phospholipids, and cholesterol arranged in a bilayer with hydrophilic surfaces enclosing a hydrophobic interior. Phospholipids have a wide range of structures but generally consist of one or more fatty acid tails linked to a polar head group that contains a phosphate. The charges of phospholipids depend on the **head groups** attached to the phosphate. Head groups can be either **neutral or negatively charged**.

Influenza A obtains its viral envelope from the plasma membrane of its host cell. However, the passage states that the composition of the envelope **differs significantly** from the plasma membrane. Unlike human membranes, the viral envelope contains mostly phosphatidylethanolamine and negatively charged glycerophospholipids. Therefore, phospholipids with **phosphoethanolamine**, glycerophospholipids with negatively charged head groups such as **phosphoserine** and **phosphoinositol**, and other negatively charged glycerophospholipids like **diphosphatidylglycerol** are needed for envelope formation. Of the choices presented, **phosphatidylserine** is the only negatively charged glycerophospholipid. Therefore, its absence is most likely to affect the assembly of the viral envelope.

(Choice A) Palmitic acid is a free fatty acid synthesized by cells. Although it can be incorporated into phospholipids, the passage does not indicate that it is required.

(Choice B) Triglycerides are storage lipids found in adipocytes (fat cells) and consist of three fatty acid tails bound to a glycerol backbone by ester linkages. They are not found in cell membranes or viral envelopes.

(Choice C) Sphingomyelin is a phospholipid with a sphingosine backbone. The passage states that *glycero*phospholipids are the major component of the viral envelope, and thus the viral lipids must contain a *glycerol* backbone.

Educational objective:
Phospholipids are composed of a hydrophilic polar head that contains a phosphate group and a hydrophobic tail with one or more fatty acids. They may contain a glycerol or a sphingosine backbone, and their properties depend on the head groups attached to the phosphate.

Time spent:0 sec
Subject:
Biochemistry

QID:403571
System:
Carbohydrates, Nucleotides, and Lipids